

When Should a Crop Map Say Less? Comparing Fixed Output Policies Under Unspecified Losses

Javier A. Rebull-Saucedo¹[0009-0008-2089-5274]* and Arthur Jafed
Zizumbo-Velasco¹[0009-0002-1603-8946]

Tecnológico de Monterrey, Mexico
rebull@exatec.tec.mx, A01796363@tec.mx

Abstract. Crop classifiers need not always emit one leaf label: a system can shrink its catalogue, abstain, return a label set, or back off to a coarser agronomic category. We compare fixed post-training policies from these four families within one seven-outcome account. Six cells carry loss; the representation is exact for losses that are constant within this predeclared basis, not for every possible downstream utility. Operating points are fixed on training and validation data, and the held-out block estimates the complete outcome-frequency vector of each policy. These vectors identify uniform dominance and linear switching boundaries without choosing a stakeholder’s loss. Across five frozen predictors on the reference-eligible PASTIS-R cohort, no within-predictor policy pair satisfied the dominance criterion over the whole declared loss simplex, while four policy types were preferred somewhere in the declared sensitivity sections. BreizhCrops supplies descriptive profiles but not paired inference under the frozen clustering unit. The study is exploratory and dataset-conditional: it does not claim geographic transport or one universally best response to uncertainty. Its contribution is a reproducible way to expose when the preferred output policy is determined by evidence and when it is determined by cost.

Keywords: Crop mapping · Selective classification · Set-valued prediction · Decision theory · Satellite image time series

1 Introduction

A parcel-level crop map is a promise. For every field in the scene it names a crop, and somebody downstream acts on that name: an inspector schedules a visit, an agronomist plans an input, an agency settles a subsidy. When the classifier is unsure, answering anyway transfers the uncertainty to that person without telling them.

There is more than one way to say less. The map can *shrink the catalogue* it is willing to promise and stop offering the classes it cannot separate; it can *abstain* on a parcel and hand it to a human; it can return a *set of candidate*

* Corresponding author.

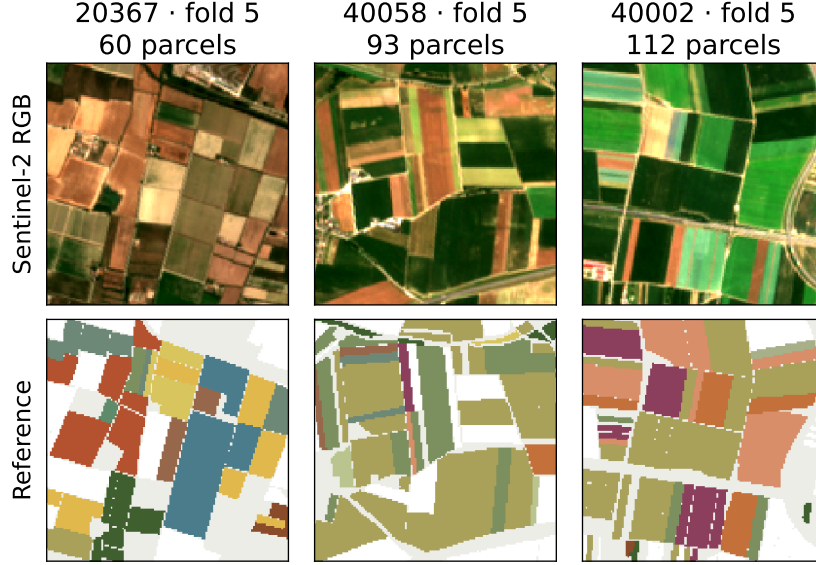


Fig. 1. Three real PASTIS-R patches in Sentinel-2 natural colour and their parcel-level reference labels. The panels were selected before this manuscript revision by a frozen density-and-class-variety rule; they provide visual context and are not model examples.

labels and let the reader pick; or it can *fall back* to a coarser agronomic category that it can defend. Each of these buys accuracy with a different currency, and each one moves the cost somewhere else.

Promising less to be wrong less often is not a new idea. The error-reject trade-off is due to Chow [5], and Ha later generalized it to selecting a *subset* of classes—together with the optimality criterion that governs it [12]. Recent work gives efficient algorithms for the subset of maximum expected utility [17] and a systematization of the available formulations [6]. In crop mapping specifically, catalogue size has already been measured as a design variable [9], and the closest work to ours rejects pixels by uncertainty on the same benchmark and on three architectures also represented in our original candidate bank, making the trade-off between precision and recall explicit [19].

What we did not find is a comparison of fixed policies from these families *against each other*. Each line of work optimizes its own device and reports its own curve. Within the search scope described in Section 2, we found no work that puts catalogue shrinking, abstention, set-valued answers, and taxonomic back-off in one decision problem, preserves the predeclared six-cell outcome basis, and asks which conclusions are invariant to every non-negative loss that is constant within those cells.

Contribution. We state the comparison and the conditions under which it can be believed. Concretely: (i) we cast fixed policies from the four families as alter-

native *actions* of the same decision problem and represent each by its complete outcome-frequency vector; (ii) we fix the estimand, the population, and the unit of analysis *before* computing anything, and we require the operating point of every mechanism to come from training and validation data alone; (iii) we derive uniform dominance and switching boundaries over the normalized six-cell loss basis, rather than choosing a favourable point; and (iv) we make the distribution of outcomes across classes a reported quantity rather than a side effect, because a mechanism that improves an average by silently dropping rare classes has not improved the same map.

What this paper does not claim. We do not infer a stakeholder’s preferences from model outputs and do not choose a loss point on their behalf. The evidence can identify uniform dominance and the boundary at which one action becomes preferable to another; it cannot say where a particular inspector, adviser, or producer lies in that space. Any later external loss point is a declared sensitivity case, not a hidden input to the primary analysis. The recorded candidate search closed without an admissible confirmatory dataset, and its dataset-specific query log was not versioned; the resulting absence claim is therefore limited to the documented candidate set rather than the whole literature. The two available parcel datasets remain exploratory. The conformal and hierarchical diagnostics reported below do not change that role and are not a completed comparison of the four actions.

2 Related Work

We organize this section by *limitation* rather than by application: for each line of work we state what it establishes and where it stops, because the stopping point is what defines the work that is left.

The renunciation framework already exists. The error–reject trade-off and its optimal rule are due to Chow [5]; that formulation is the degenerate case in which the only alternative to answering is silence. Ha generalized it to class-selective rejection, showing that selecting a subset of classes is the optimal compromise between error and the mean number of classes returned [12]; the treatment is decision-theoretic and assumes the posteriors are known. Ha’s class-selective rule is an implemented parameterized literature family. It was not part of the already executed six-arm family; a later prospective exploratory extension freezes one validation-derived member of that family as a seventh arm, without turning it into a confirmatory comparison. Mortier et al. give algorithms for the Bayes-optimal subset of maximum expected utility, with a utility that is a function of set size [17], and Chzhen et al. unify the existing set-valued formulations and their trade-offs [6], at infinite sample and under a plug-in principle. The framework is therefore not our contribution, and we say so on the first page. What these treatments share is that the price of an answer is a function of *how large* the answer is. They do not compare fixed policies from the four delivery families through the same declared six-cell outcome basis.

Learned abstention is developed, and out of our scope by choice. Deep selective classification supplies risk guarantees [7], an architecture with an integrated reject head [8], a wagering-style loss [15], and an AUC-based selection criterion [18]. All of them abstain *per instance*; none emits a set or falls back to a coarser class. We do not reimplement them and we do not benchmark against them: our question is how abstention compares with the other three actions across a common loss space, not whether a better abstention rule exists.

Conformal and reward-based sets fix a target we deliberately leave open. Adaptive conformal classification attains marginal coverage with approximate conditional coverage through a new conformity score [20]; regularizing the scores of unlikely classes yields sets five to ten times smaller at the same coverage [1]. Both fix a coverage target and treat set size as a consequence or a secondary objective. Risk-controlling prediction sets instead require a loss and a risk level, bounding an arbitrary user-specified loss with finite-sample guarantees from a holdout [3]. Partial reject options presuppose a reward: set size itself becomes the decision, from outlier through firm answer to full rejection, optimal under a given reward with size-based penalties [14]. These methods fix either a coverage target or a loss specification within one output family. We instead leave the six-cell loss weights unspecified and report where preference changes across output families.

Renunciation has reached Earth observation, and not through agriculture. The closest work to ours rejects pixels by an uncertainty threshold on PASTIS, using three architectures represented in our original candidate bank, and makes the trade-off explicit: not answering lowers recall and can raise precision when the rejected pixel would have been misclassified [19]. That is the risk–coverage compromise, performed and acknowledged. What it does not do is place the decision in the selective-classification frame, compare alternative actions, or characterize the loss-space boundaries at which the preferred action changes. Elsewhere in Earth observation, SHRUG-FM reports abstention with interpretable thresholds [11], on burn scars, floods, and landslides rather than on any agricultural task. Rejection methods have been applied to vegetation mapping [10], on airborne hyperspectral data rather than per-parcel satellite time series. Open-set crop recognition rejects the *unknown* through outlier exposure [4], which is a different question from distributing a degree of promise among known classes.

Catalogue size is already a measured design variable. Ghassemi et al. treat the number of classes as a design variable and measure it: starting from 52 LUCAS classes, a 26-class scheme balances accuracy against detail [9]. This is the direct precedent for reading catalogue shrinking as an operating point, and it forecloses any claim that we are the first to do so. There the setting is land cover with Random Forest; here it is one policy family among four, represented in a common outcome vocabulary over the loss class induced by the declared six-cell basis.

Taxonomic back-off already has a crop-mapping precedent. Multi-scale label hierarchies improve crop mapping from satellite image time series [23]. That work

both predicts at three granularities and allows the output granularity to follow classification confidence; it is therefore a direct precedent for confidence-driven taxonomic back-off, not merely for hierarchical training. Hierarchical fusion also combines satellite, rotational, and contextual signals [2], but for information fusion rather than variable-granularity delivery. We do not claim the back-off mechanism as new. Hierarchical set-valued prediction has also been formalized with internal nodes as the usual valid sets [16]. The open comparison is whether this action is uniformly preferable to another action, and where their preference boundary lies otherwise.

Distributional effects are documented outside our domain. Selective classification can *magnify* disparities between groups: abstaining more can make outcomes worse for some of them [13]. The evidence is on demographic groups in vision and language, not on agronomic classes or parcels. We take it as the reason to report distribution across classes rather than as evidence about crops, and we note that the result runs opposite to the intuition that declining to answer is a neutral act.

Novelty, stated with its universe. We found no work that compares catalogue shrinking, per-parcel abstention, set-valued prediction, and taxonomic back-off *as alternative actions of the same decision problem*, reports their uniform-dominance order over one predeclared six-cell loss basis, and accounts for how the underlying outcomes are distributed across classes. This statement is bounded by our search: arXiv, OpenAlex, and Semantic Scholar, 61 queries plus 6 manual ones, covering 2019–2026 together with the classics retrieved by name, in English only. That procedure indexes remote-sensing proceedings without DOIs poorly, and it is written in the vocabulary of selective classification—a limitation we observed in practice, since the closest work to ours [19] uses none of those terms and was found by another route.

3 Problem Statement and Estimand

3.1 Four Ways to Promise Less

Let $\mathcal{Y}$ be the class catalogue, let x_i be the observations for parcel i , and let $y_i \in \mathcal{Y}$ be its reference label. A conventional classifier commits to a map $x_i \mapsto \hat{y}_i \in \mathcal{Y}$: it must name exactly one class for every parcel, whatever its posterior looks like. The mechanisms compared here all relax that commitment in the same formal way. Each replaces the point prediction with a subset of the catalogue,

$$C_m(x_i) \subseteq \mathcal{Y}, \quad C_m(x_i) = \emptyset \text{ permitted}, \quad (1)$$

and they differ only in which subsets they may emit, and in what fixes the choice.

Catalogue shrinking. A subcatalogue $\mathcal{S} \subset \mathcal{Y}$ is fixed before any parcel is seen. A parcel whose unrestricted posterior maximizer lies inside $\mathcal{S}$ receives the restricted maximizer; a parcel whose maximizer lies outside receives nothing. The rule reads the posterior and never the reference label—keying delivery on the truth would make the mechanism unimplementable at inference time, and an earlier implementation in this line of work did exactly that. The decision is per class, and it is taken once.

Per-parcel abstention. The mechanism emits a singleton or the empty set, according to whether a confidence score clears an operating point. The decision is per parcel, and it is taken as many times as there are parcels.

Set-valued prediction. The mechanism emits a non-empty subset whose size varies with the difficulty of the parcel and is calibrated to a nominal marginal-containment target. Any finite-sample guarantee additionally requires exchangeability between calibration and evaluation; we report observed containment and do not claim that spatial resampling repairs that assumption. Every parcel receives an answer; what varies is how much the answer commits to.

Taxonomic back-off. The mechanism emits the set of leaves under a node of a class hierarchy. Formally this is a subset like any other; what distinguishes it is that the admissible subsets are those an agronomist can name, so the answer degrades to a coarser but still communicable category instead of to a list.

Two degenerate elements of the same space bound it: full-catalogue singleton prediction, which renounces nothing, and universal non-delivery, for which $C \equiv \emptyset$. Neither is one of the mechanisms, because neither chooses what to give up. They are not on the same footing, however, and Section 4 keeps them apart: the unmodified singleton is evaluated, as the reference against which every renunciation is priced, whereas universal non-delivery only fixes the price of silence and is never run. Writing all of these in one action space is this paper’s first move, and it is what makes the question askable at all: as long as legend design, thresholding, conformal calibration, and hierarchical prediction are treated as separate techniques, there is no object on which to ask which of them one should prefer.

3.2 The Estimand

Fix a dataset d with its region, campaign, split, and predictor, and a policy m operating at τ_m . Its full realized object is $Z_{i,m} = (y_i, C_m(x_i), a_m)$, where a_m records whether a non-singleton answer is an arbitrary label set or a named taxonomic category. A predeclared coarsening $\pi(Z)$ maps that object to one cell of a closed outcome account: correct precise delivery, incorrect precise delivery, non-delivery, a set that contains or excludes the truth, or a coarse answer that contains or excludes the truth. Correct precise delivery is the zero-loss reference R_0 ; write the remaining cells as $L_1, \dots, L_6$. The empirical quantity is their

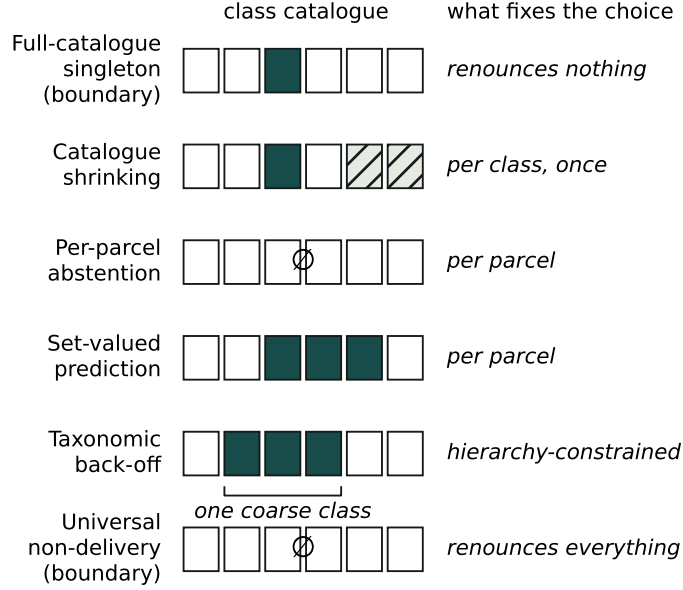


Fig. 2. The evaluated policies share one output type: a subset of the leaf catalogue, with the empty set allowed. The diagram is structural; its cells do not represent classes or measured frequencies.

frequency vector,

$$q_{d,m,j} = \frac{1}{N_d} \sum_{i=1}^{N_d} \mathbf{1}\{O_i(m) = L_j\}, \quad j = 1, \dots, 6, \quad (2)$$

where N_d counts *all* eligible parcels of that block. For any finite non-negative loss that is constant within the six cost-bearing cells of π , common positive scale is irrelevant to the ordering, so we normalize it to

$$\ell \in \Delta^5 = \left\{ \ell \in \mathbb{R}_+^6 : \sum_{j=1}^6 \ell_j = 1 \right\}, \quad R_{d,m}(\ell) = \mathbf{q}_{d,m}^\top \ell. \quad (3)$$

This representation is exact for the declared loss class $L_\ell(Z) = \ell_{\pi(Z)}$, with zero assigned to R_0 . It is not complete for every downstream utility: for example, two truth-containing sets of different sizes occupy the same cell, as do two wrong crop labels with different consequences. Uniformity below is therefore with respect to the six-cell basis, not to every conceivable use of a crop map.

The data determine $\mathbf{q}$, not ℓ . We therefore report the full linear decision geometry instead of inserting a point. Arm m uniformly dominates arm n exactly

when $q_{m,j} \leq q_{n,j}$ for every cost-bearing cell and at least one inequality is strict. Otherwise, their switching boundary is the hyperplane $(\mathbf{q}_m - \mathbf{q}_n)^\top \boldsymbol{\ell} = 0$. A scalar switching price exists only inside a declared section that fixes the other coordinates. A two-dimensional picture is consequently a sensitivity view, not the full simplex induced by the declared basis.

Cardinality is not the loss. Expected set size prices two sets of equal size identically even when one excludes the truth, and it assigns zero size to non-delivery. We retain it as a descriptor but never use it as the common objective. Nor do we infer a loss point from the model outputs: that would turn a modeling preference into a data result. A later externally sourced point may be overlaid as a non-confirmatory sensitivity case, with its provenance stated.

The population includes the parcels that receive nothing. N_d ranges over every eligible parcel in the block, including those the mechanism declines to answer and those whose class the shrunken catalogue no longer contains. Restricting the denominator to delivered parcels would price renunciation at zero and would score each mechanism on a population of its own choosing, so that a mechanism could improve its reported quality by answering less.

The operating point is fixed before the held-out block is touched. τ_m is derived entirely from training and validation. The held-out block estimates the realized outcome and nothing else. Re-equalizing loss, coverage, or delivery rate on the held-out block is prohibited, including as a post-hoc alignment intended to make mechanisms comparable: it is the leak that two audits of an earlier version of this work found, first in the threshold and then in its target rate.

3.3 Scope, Dependence, and What Is Not Claimed

Inference is *conditional on each dataset*, with its region, campaign, split, and predictor. The observational unit is the parcel. Parcels are not independent: they arrive in image patches that share acquisition, geography, and annotation, so the dependence cluster is the patch. The inferential bootstrap therefore resamples whole patches within spatial blocks and never parcels one by one. The primary PASTIS-R atlas treats its held-out outer fold as one stratum and resamples the patches within it. A separate sensitivity repartitions those same whole patches at every declared k and reruns the complete simultaneous profile family; it neither drops parcels nor turns subdivisions into independent territories. Below three unique paired clusters, a profile is reported descriptively, without an inferential claim, because at that size the interval would describe the resampling scheme rather than the data.

No quantity is pooled across datasets. Each dataset is reported on its own; the second supplies a descriptive external-region contrast rather than an additional sample of the first. The number of spatial blocks is a sensitivity parameter for grouping the same evidence, not a count of replicates and not a lever on statistical power.

What this design gives up is stated here rather than in a limitations paragraph, because it constrains every claim that follows: **we do not claim transport**. Nothing reported for one region or campaign is asserted to hold in another. A conditional estimand is what the available partitions support; a marginal one would require a sample of regions that this work does not have, and the two were mixed in the design of an earlier version until an audit separated them.

No deployment geography is part of the scientific claim. The evaluated regions support dataset-conditional statements only; applying the policy comparison elsewhere requires new reference data and a prospectively declared evaluation design.

4 Fixed Policies as Alternative Actions

The main analysis instantiates fixed points from four post-training policy families as six arms: a baseline, two catalogue rankings, abstention, set prediction, and taxonomic back-off. They do not cover the lattice. Cascades alter information and computation; class-specific thresholds add sparse validation-tuned parameters; learned rejectors alter training. Learned, class-adaptive, and multi-stage policies remain unexplored, so claims are restricted to these four points and the two catalogue rankings. Universal non-delivery is a boundary. Per model-stratum, Ha’s prospective exploratory extension fixes its cost at the smallest second-highest validation posterior attaining the singleton target, without held-out labels or rematching; absent a confirmatory bank, it is diagnostic rather than an admission test.

4.1 A Common Output Type

Every arm returns the Boolean membership vector of $C_m(x_i)$ over the fixed catalogue. The representation contains no reference label, loss, or delivered-only filter. A row with one true entry is a precise delivery, a row with several true entries is an ambiguous or coarsened delivery, and an all-false row is non-delivery. Keeping the empty set inside the output type is important: it ensures that an abstained parcel remains in the population rather than disappearing before it can be counted.

The baseline emits the singleton containing the posterior maximizer. Both catalogue-shrinking arms apply the same operator to a different predeclared subcatalogue. If the unrestricted posterior maximizer belongs to that subcatalogue, it is emitted; otherwise the output is empty. The operator does not inspect the reference label. The two arms therefore differ only in the training-side rule that orders classes for withdrawal, not in how a test parcel is handled.

Per-parcel abstention emits the posterior maximizer when its confidence clears a fixed threshold and the empty set otherwise. Set-valued prediction sorts labels by posterior probability and emits the smallest non-empty prefix reaching a calibrated cumulative mass. The constructor applies that mass but never estimates it: calibration belongs entirely to training and validation, and any coverage

guarantee depends on that preceding calibration rather than on the constructor alone.

The set-valued arm uses deterministic, non-randomized split-conformal adaptive prediction sets. For each calibration parcel, the score is the cumulative posterior mass through the true label under a stable probability ranking. The calibrated mass is selected by the exact finite-sample order-statistic rank $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$, not by passing a rank fraction to an interpolating quantile API. If that rank lies beyond the calibration sample, the arm emits the complete catalogue rather than substituting the largest observed score. We choose the deterministic convention to avoid introducing an auxiliary randomization and seed; its possible conservatism is reported through observed marginal containment and set size. The usual finite-sample statement is conditional on exchangeability; spatially grouped parcels do not make that assumption automatic. Held-out coverage is therefore reported as empirical containment over all eligible parcels and is never used to recalibrate the mass.

Taxonomic back-off uses a predeclared fourth-edition HCAT cut over the leaf catalogue. A confident parcel keeps its singleton; an uncertain parcel receives every represented leaf below the node containing the posterior maximizer. The committed mapping resolves every leaf and cut node against the EuroCrops taxonomy before constructing a prediction, and rejects missing, repeated, non-ancestral, or out-of-catalogue assignments. Three cut nodes contain only one represented study leaf; back-off in those cases leaves the visible singleton unchanged rather than jumping post hoc to a broader node. Consequently, a malformed hierarchy cannot silently turn a parcel into non-delivery, and a deeper hierarchy would be a different operating rule rather than a choice made after observing outcomes.

All operating points—the retained catalogue, confidence threshold, calibrated mass, and coarse partition—are inputs to these operators. None is fitted by them, and none may be re-equalized on the held-out block. This separation is what makes test-label invariance inspectable: changing the test labels cannot change a delivered set.

Invariance to the held-out labels is a property of the operators; disjointness of the blocks is a property of whoever calls them, and the implementation no longer assumes it. A fitted operating point carries the stable parcel identities of the rows used to fit it and refuses to be applied to any of them. Identity is not inferred from posterior bytes: distinct parcels may have identical posteriors, and recomputing the same parcel may change its bytes. Handing the evaluated block back as calibration data therefore fails rather than passing quietly, which is how the corresponding leak entered an earlier version of this work.

The implementation makes this temporal order part of the interface. Its fitting function accepts training and validation arrays but has no held-out argument; its application function accepts held-out posteriors but has no label argument; only the evaluator receives labels, after all six membership matrices exist. A single orchestration path then composes these stages without supplying a default catalogue size, delivery target, miscoverage level, back-off rate, hierar-

chy, or loss point. This prevents an absent preregistration choice from becoming a library default.

The leaf catalogue is fixed by the training-side model output. Separately, a neutral evaluation universe is selected once from training labels using the predeclared support floor. Validation and held-out labels cannot add or remove a class from that universe. Every arm retains every held-out parcel in its outcome denominator, including non-deliveries and parcels whose true class lies outside the neutral classwise summary. A selected class absent from a held-out block remains visible with zero support and an undefined classwise statistic; it is not silently dropped or converted to a valid zero. Thus every primary frequency and classwise diagnostic uses denominators fixed before the arm’s delivery is observed.

4.2 The Declared Action–Outcome Basis

The outcome interface follows the closed coarsening fixed by the loss-space contract. A precise label is either correct, the zero-loss reference, or wrong. An empty set is non-response. A multi-label set either contains or excludes the truth. A coarse class likewise either contains or excludes the truth. These are seven cells in total. The two failure cells for non-singleton answers are indispensable: without them, a prediction set or a coarse answer would be charged only when it succeeds and would again acquire a free failure mode.

Classification into a cell follows the delivery visible to the user, not merely the name of the algorithm. A singleton emitted by the set-valued arm is scored as a precise label. A multi-label delivery is distinguished as an unconstrained label set or a named coarse class, because the two may have different consequences. An empty row is non-response under every arm. Every parcel maps to exactly one cell.

The primary evaluator counts all seven cells and returns their complete frequency vector without accepting a loss point. A separate sensitivity interface accepts six non-negative finite losses, requires the correct precise label to remain the zero reference, and refuses an incomplete mapping. After removing common scale, these points form a five-dimensional simplex. Synthetic points test the necessary property that changing the relative price of a wrong label and silence can reverse which action has lower risk; they do not become empirical preferences. Completeness refers to this account: utilities that vary within a cell need a richer basis and are outside the reported dominance claim.

Every cell frequency uses the same eligible-parcel denominator in Eq. (2), including non-deliveries. Risk at any declared sensitivity point is the dot product in Eq. (3). Expected set size remains descriptive; the earlier size-based utility links diagnostically to macro recall but does not rank the six arms.

Four crops fill most of the mosaic
one cell = one parcel; schematic position

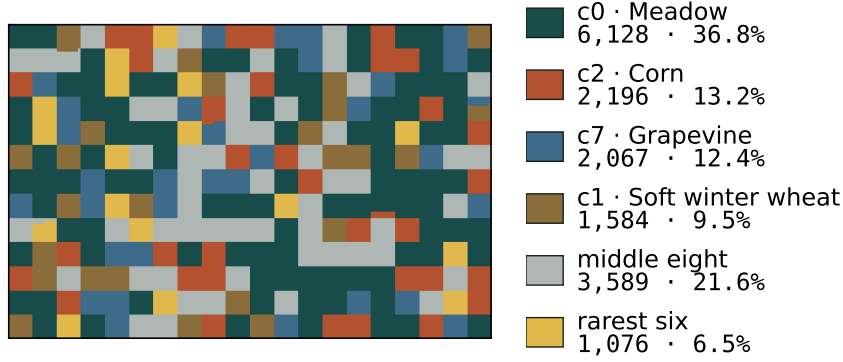


Fig. 3. Exact class support in the reference-eligible PASTIS-R fold-five cohort. Each micro-cell represents one parcel; only its count is literal. Position and shape carry no geographic or area meaning. The four leading classes are shown separately, followed by the middle eight and rarest six.

5 Evaluation Protocol

The protocol first estimates each arm’s complete outcome-frequency vector, then derives dominance and switching geometry over every normalized loss. It needs no point loss, and no empirical contrast was run to write this section.

5.1 Split Order and Pair Formation

The empirical banks are the optical modality of PASTIS-R [22] and BreizhCrops [21]. Naming PASTIS-R identifies the distributed bank; no radar channel enters the evaluated predictors.

The reference-eligible PASTIS-R fold-five cohort contains 16640 parcels. Its four most frequent classes account for 72.0% of parcels, whereas the six rarest account for 6.5%. These are parcel shares, not mapped-area shares; the mosaic in Fig. 3 is deliberately schematic rather than geographic.

Training fixes the neutral class universe and support-based catalogue, validation fixes the other operating-point components, and evaluation starts only after both are immutable. Stable parcel identities enforce disjoint splits and prevent applying a fitted point to any training or validation identity, including a copied row; posterior equality is irrelevant.

The common catalogue size is the smallest prefix of classes ordered by training support whose cumulative training mass reaches 0.9, with class id breaking support ties; the F-score and support rules use that same size. Validation targets

Table 1. Fixed output policies evaluated in the primary six-arm comparison. The table describes policy instances, not the full family of possible operating points.

Policy	Fixed before evaluation	Visible response
Singleton baseline	Posterior argmax	One leaf label
Support catalogue	Smallest training-support prefix reaching 0.9 mass	One label or silence
F-score catalogue	Same catalogue size, alter-native frozen ordering	One label or silence
Abstention	Validation threshold target-ing 0.9 delivery	One label or silence
Adaptive set	Split-conformal mass at $\alpha = 0.1$	Non-empty label set
HCAT4 back-off	Validation threshold target-ing 0.9 precise leaves	Leaf or named coarse class

a 0.9 delivery rate for abstention, a 0.1 miscoverage level for the adaptive set, and a 0.9 precise-leaf rate for taxonomic backoff. A class enters the neutral universe with at least 20 training parcels. These are author-chosen nominal intervention points, not equal-loss points, and none may be rematched on the held-out split.

The evaluator assigns every eligible held-out parcel under every arm to one of the seven closed outcomes. Pairing is by identity, with matching block and `patch_id`, so row reordering is allowed but omission, duplication, and substitution are not. The training-derived class universe travels with the profile and is never reconstructed from evaluation or a bootstrap draw.

Both banks had been inspected before this execution and remain exploratory. A closed screening register admitted no separate confirmatory dataset; this is a bounded result over the recorded candidates, not an exhaustive claim about public data.

5.2 Two Declared Uncertainty Views

For each arm pair and cost cell, the contrast is the left-arm indicator minus its comparator. The primary view resamples whole `patch_id` groups within blocks, carries every parcel and arm together, and combines blocks using their original parcel shares. There is no parcel bootstrap.

Separately, indicator differences are averaged within spatial blocks and those means receive equal weight. A paired Student interval over a declared coordinate or section exposes between-block variation without treating overlapping folds as independent regions. Its centre may differ from the parcel-population contrast when block sizes differ; forcing equality would change the estimand.

Both views retain the complete denominator. Below three unique paired units, the implementation returns deltas and a reason but no interval, p -value, or Holm eligibility. A comparison declared identical by construction is retained as a self-check and excluded from multiplicity.

5.3 Dominance, Boundaries, and Multiplicity

Every stratum contains all unordered arm pairs and all six cost-bearing outcome coordinates. The primary scalar for an ordered pair is

$$D_{m,n} = \max_{j=1,\dots,6} (q_{m,j} - q_{n,j}). \quad (4)$$

Observed componentwise dominance is descriptive. An inferential dominance claim requires every simultaneous one-sided coordinate bound to be at most zero and at least one to be strictly below zero. The bounds are obtained from a maximum statistic over the complete predeclared family. Pairs or coordinates are not removed after their estimates are seen. If a required comparison is missing or has too few paired clusters, the affected simultaneous claim is withheld and the observed vectors remain available as descriptive output.

The frozen primary procedure uses a family-wise alpha of 0.05, 95% intervals, 9,999 paired patch-cluster resamples within spatial block, add-one bootstrap probabilities, and seed 20260909. Within each confirmatory stratum, the critical value is the maximum absolute studentized centred deviation over all 15 unordered arm pairs and 6 cost coordinates: 90 coordinates, or 180 directional inequalities. A zero standard error withholds the inferential claim, except where neither action can produce that outcome: their difference is exactly zero in every cluster, so that bound is the exact zero and carries no bootstrap probability. This structural bound is load-bearing: without it that pair cannot retain six bounded coordinates. We recorded this clarification after execution and re-derived both affected outputs before submission. Studentization is performed inside every resample with that resample's own block-centred patch-cluster standard error; reusing one denominator for all resamples is prohibited. A relation stated across predictors uses an intersection-union rule and must hold in the same direction in every predeclared predictor stratum; no favourable predictor is selected.

The fitted winner region of arm m is the intersection of the simplex with the half-spaces $(\mathbf{q}_m - \mathbf{q}_n)^\top \boldsymbol{\ell} \leq 0$ for every comparator n . The boundary for one pair is the corresponding equality. These objects use all coordinates and do not require a probability measure over the simplex. A plotted two-dimensional section must declare the coordinates it fixes; it cannot be labelled as the full map or converted into a universal scalar threshold.

Point losses are limited to non-confirmatory sensitivity cases with an external source or declared geometric section. They cannot select the operating point, define the primary family, or support a stakeholder claim. Earlier ratios and Fieller sets remain outside the primary analysis.

6 Results

We report exploratory outcome accounting, not a universal preference. Figure 4 shows the empirical object behind every comparison before any loss is applied. It makes the trade visible: abstention moves mass from wrong precise labels to silence; adaptive sets move most mass to truth-containing ambiguous answers;

back-off introduces both correct and incorrect coarse deliveries. None of those moves has a sign until its cells receive a loss.

For the 5 frozen PASTIS-R predictors, the primary atlas certified no dominance: 71 of 75 action pairs were certainly incomparable and 4 undetermined, not tied. This absence of dominance held in all ten seed runs. The historical finding that no spatial partition was selected failed in every seed run and is withdrawn; lacking per-seed selection profiles, we report no replacement.

No preregistered loss point exists to mark: point-loss selection is prohibited. Across the declared planar sections, the singleton baseline, adaptive sets, abstention, and taxonomic back-off each won somewhere; neither catalogue policy won a sampled point. This establishes switching on the declared grid, not a probability of being best or a volume of the simplex.

Figure 5 shows the first section of the frozen exhaustive enumeration, with $L_4 = L_5 = L_6 = 0.1$ and no pooling across predictors. It was selected for display after materialization, not preregistered. The 20 declared two-dimensional sections contain 66 lattice points each. All 20 sections remain available in the structured artifact.

BreizhCrops does not contribute a paired inferential atlas. Each model–region stratum supplies 1 cluster under the frozen unit, below the minimum of 3, so no simultaneous interval, dominance claim, or phase boundary is estimable. Its outcome profiles remain descriptive; the dataset is not pooled with PASTIS-R to manufacture replication.

The conformal mass and the HCAT4 precise-delivery threshold were fitted on the frozen validation split. Both deliveries were fixed before evaluation labels were opened; reference availability only restricted the already-fixed PASTIS-R deliveries. Their validation targets are different operating constraints, not an equal loss.

Every conformal row met its marginal containment target, but only by returning multi-label answers. HCAT4 recovered additional truth-containing deliveries beyond correct precise answers in every row, yet also emitted wrong coarse answers in every row. Its realized precise-delivery rate varied across held-out strata despite one validation target. Thus a named coarse answer is neither a free rescue nor the same object as an arbitrary truth-containing set.

Marginal conformal calibration did not imply class-conditional coverage: one PASTIS-R U-Net class contained the truth in 28 of 206 sets. The historical extreme BreizhCrops support was 6 parcels; seed-grid extrema spanned 1–6 and cannot sustain a class-level generalization.

Rows use reference-eligible evaluation cohorts: 16640 valid-label PASTIS-R parcels and 30000 BreizhCrops parcels per external-region stratum. The other 11892 PASTIS-R posterior identities lack a valid semantic reference for this outcome account. Delivery was fixed before that reference filter was applied, so the filter cannot improve a policy’s delivery rate; those identities are not silently described as evaluated parcels.

DeepLabv3+							
singleton	64.9	35.1	0.0	0.0	0.0	0.0	0.0
support catalogue	63.4	31.7	4.8	0.0	0.0	0.0	0.0
F1 catalogue	63.5	30.7	5.8	0.0	0.0	0.0	0.0
abstention	62.2	26.7	11.1	0.0	0.0	0.0	0.0
label set	5.8	0.3	0.0	91.1	2.8	0.0	0.0
back-off	62.9	28.5	0.0	0.0	0.0	4.3	4.3
SegFormer							
singleton	64.7	35.3	0.0	0.0	0.0	0.0	0.0
support catalogue	63.5	33.0	3.5	0.0	0.0	0.0	0.0
F1 catalogue	63.2	32.0	4.7	0.0	0.0	0.0	0.0
abstention	61.7	27.0	11.3	0.0	0.0	0.0	0.0
label set	5.5	0.1	0.0	92.2	2.2	0.0	0.0
back-off	62.3	28.5	0.0	0.0	0.0	5.3	3.9
TSViT-Pheno							
singleton	85.8	14.2	0.0	0.0	0.0	0.0	0.0
support catalogue	79.9	12.4	7.7	0.0	0.0	0.0	0.0
F1 catalogue	77.4	10.5	12.1	0.0	0.0	0.0	0.0
abstention	80.6	7.6	11.8	0.0	0.0	0.0	0.0
label set	6.8	0.1	0.0	91.6	1.5	0.0	0.0
back-off	82.3	9.5	0.0	0.0	0.0	5.0	3.2
TSViT-FullIM							
singleton	88.1	11.9	0.0	0.0	0.0	0.0	0.0
support catalogue	81.4	10.2	8.4	0.0	0.0	0.0	0.0
F1 catalogue	77.1	7.8	15.1	0.0	0.0	0.0	0.0
abstention	81.9	5.5	12.6	0.0	0.0	0.0	0.0
label set	6.7	0.1	0.0	91.9	1.3	0.0	0.0
back-off	84.1	7.3	0.0	0.0	0.0	5.4	3.2
U-Net							
singleton	55.4	44.6	0.0	0.0	0.0	0.0	0.0
support catalogue	55.1	43.8	1.1	0.0	0.0	0.0	0.0
F1 catalogue	55.3	43.9	0.8	0.0	0.0	0.0	0.0
abstention	52.5	35.3	12.2	0.0	0.0	0.0	0.0
label set	2.7	0.3	0.0	90.9	6.1	0.0	0.0
back-off	53.8	38.5	0.0	0.0	0.0	3.4	4.3
	R0	L1	L2	L3	L4	L5	L6
	correct						

Fig. 4. Complete PASTIS-R outcome-frequency profiles for every frozen predictor and primary policy. Cell labels are percentages of the same reference-eligible cohort. Shade repeats the printed magnitude and is not the sole information channel. Columns are the zero-loss reference followed by the six cost-bearing cells defined in Section 3.2.

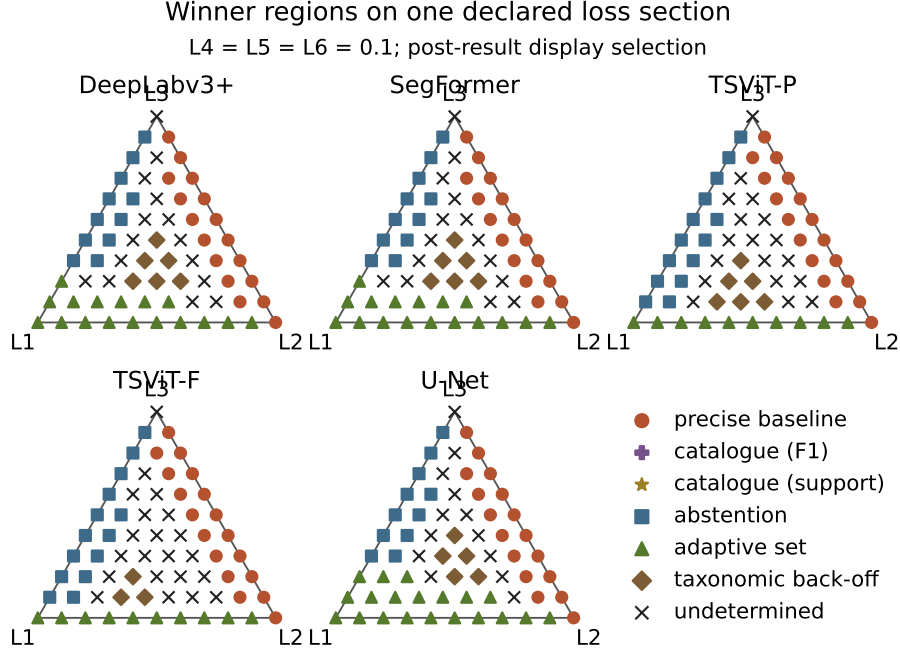


Fig. 5. Certain winner regions on one declared section of the exploratory PASTIS-R loss atlas. Symbols identify the winning action; crosses are evaluations for which the simultaneous band does not establish one winner. Each panel is a frozen predictor. Catalogue-restriction actions remain in the legend even though neither wins on this section, closing the six-action account. The displayed section is the first in the frozen enumeration and was selected for presentation after results.

Together, the profiles and atlas answer the narrower question supported by the data: which policy is preferred for a declared six-cell loss, and where that preference changes. They do not identify the loss itself.

7 Discussion

The diagnostics expose why one success rate cannot answer the decision question. Conformal sets attain marginal containment by returning multiple labels; HCAT4 sometimes recovers the truth through a communicable parent and sometimes returns the wrong parent. Calling either uniformly better would silently price breadth or coarse error at zero, so the outcome coordinates must remain separate.

The coverage statement is conditional on each evaluated bank and on exchangeability between its calibration and evaluation parcels. Both banks had been inspected before this execution, so every result here is exploratory even

though the evaluation partitions are held out from calibration. No transport claim follows, and the closed dataset search did not find a separate admissible confirmatory benchmark. Predictor-to-predictor variation is sensitivity, not a route to select a favourable model. PASTIS-R therefore supports an exploratory decision atlas; BreizhCrops supports descriptive outcome profiles but not paired inference because its frozen reporting structure has only 1 cluster per model–region stratum.

No certified dominance is not a tie: crossing profiles and wins for four actions reject a price-free ranking while locating its switches. Catalogue restriction winning no sampled point does not prove an empty winner region because the grid has no simplex measure. The spatial grouping conclusion changed under the frozen seed grid, so we withdraw it rather than choose a favourable partition after seeing the atlas.

Two scope restrictions are load-bearing. First, each arm is one fixed operating policy; the study does not order every possible member of its family. Second, the six-cell account intentionally forgets set cardinality, taxonomic depth, and which crop pair was confused. A downstream use that prices any of those distinctions requires an expanded outcome basis before applying the same geometry.

The inherited PASTIS-R checkpoints do not record training seeds in their declared identity sources. Seed sensitivity can therefore rerun only post-training stochastic stages for those members; it cannot establish robustness to training randomness. We do not backfill that claim by retraining after seeing the atlas. Stable predicates are retained; changed counts above are ranges. BreizhCrops sampling and classical fits remain rerunnable and separate, without pooling.

Label-noise robustness is likewise unmeasured: no transition model tailored to either bank was declared, and the withheld secondary family supplies no estimable corruption target. We therefore do not inject a convenient synthetic noise process after seeing the results.

Data and Code Availability

The anonymous source archive contains the L^AT_EX project, bibliography, generated appendix, and figures, but no raw satellite data. A separate anonymous supplement reconstructs all seventy-five pairwise atlas verdicts and the declared lattice from aggregate patch counts; it contains no imagery, parcel records, repository access, or author metadata. Training artifacts remain private during review and will accompany a citable archive before publication. Source data retain their providers’ terms; readers re-run the atlas and figures from outcome counts, not imagery.

References

1. Angelopoulos, A., Bates, S., Malik, J., Jordan, M.I.: Uncertainty sets for image classifiers using conformal prediction (2020)

2. Barriere, V., Claverie, M., Schneider, M., Lemoine, G., d'Andrimont, R.: Boosting crop classification by hierarchically fusing satellite, rotational, and contextual data. *Remote Sensing of Environment* **305**, 114110 (2024). <https://doi.org/10.1016/j.rse.2024.114110>
3. Bates, S., Angelopoulos, A., Lei, L., Malik, J., Jordan, M.: Distribution-free, risk-controlling prediction sets. *Journal of the ACM* (2021). <https://doi.org/10.1145/3478535>
4. Carvalho, T., Martinez, J.A.C., Oliveira, H., dos Santos, J.A., Feitosa, R.Q.: Outlier exposure for open set crop recognition from multitemporal image sequences. *IEEE Geoscience and Remote Sensing Letters* **20**, 1–5 (2023). <https://doi.org/10.1109/LGRS.2023.3244532>
5. Chow, C.: On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory* (1970). <https://doi.org/10.1109/tit.1970.1054406>
6. Chzhen, E., Denis, C., Hebiri, M., Lorieul, T.: Set-valued classification — overview via a unified framework (2021)
7. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. pp. 4878–4887. Curran Associates (2017), <https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html>
8. Geifman, Y., El-Yaniv, R.: SelectiveNet: A deep neural network with an integrated reject option. In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*. *Proceedings of Machine Learning Research*, vol. 97, pp. 2151–2159. PMLR (2019), <https://proceedings.mlr.press/v97/geifman19a.html>
9. Ghassemi, B., Izquierdo-Verdiguier, E., d'Andrimont, R., Vuolo, F.: How accurately and in what detail can land use and land cover be mapped using copernicus Sentinel and LUCAS 2022 data? *Remote Sensing* **17**(8), 1379 (2025). <https://doi.org/10.3390/rs17081379>
10. Gimenez, R., Laloue, A., Fabre, S.: Rejection methods for vegetation mapping using hyperspectral airborne data. *International Journal of Remote Sensing* **44**(16), 4937–4962 (2023). <https://doi.org/10.1080/01431161.2023.2240520>
11. Gonzalez-Calabuig, M., Cohrs, K.H., Nedungadi, V., Osika, Z., Cartuyvels, R., Knoblauch, S., Massant, J., Nath, S., Ebel, P., Sitokoustantinou, V.: SHRUG-FM: Reliability-aware foundation models for Earth observation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. pp. 7919–7928 (2026)
12. Ha, T.M.: The optimum class-selective rejection rule. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **19**(6), 608–615 (1997). <https://doi.org/10.1109/34.601248>
13. Jones, E., Sagawa, S., Koh, P.W., Kumar, A., Liang, P.: Selective classification can magnify disparities across groups. In: *9th International Conference on Learning Representations (ICLR)* (2021), https://openreview.net/forum?id=NOM_4BkQ05i
14. Karlsson, M., Hössjer, O.: Classification under partial reject options. *Journal of Classification* (2023). <https://doi.org/10.1007/s00357-023-09455-x>
15. Liu Ziyin, Wang, Z., Liang, P.P., Salakhutdinov, R., Morency, L.P., Ueda, M.: Deep gamblers: Learning to abstain with portfolio theory. In: *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*. pp. 10622–10632. Curran Associates (2019), <https://proceedings.neurips.cc/paper/2019/hash/0c4b1eeb45c90b52bfb9d07943d855ab-Abstract.html>

16. Mortier, T., Hüllermeier, E., Dembczyński, K., Waegeman, W.: Set-valued prediction in hierarchical classification with constrained representation complexity. In: Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence. Proceedings of Machine Learning Research, vol. 180, pp. 1392–1401. PMLR (2022), <https://proceedings.mlr.press/v180/mortier22a.html>
17. Mortier, T., Wydmuch, M., Dembczyński, K., Hüllermeier, E., Waegeman, W.: Efficient set-valued prediction in multi-class classification. *Data Mining and Knowledge Discovery* **35**, 1435–1469 (2021). <https://doi.org/10.1007/s10618-021-00751-x>
18. Pugnana, A., Ruggieri, S.: AUC-based selective classification. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS). Proceedings of Machine Learning Research, vol. 206, pp. 2494–2514. PMLR (2023), <https://proceedings.mlr.press/v206/pugnana23a.html>
19. Rey, M., Mnih, A., Neumann, M., Overlan, M., Purves, D.: Uncertainty evaluation of segmentation models for Earth observation (2025)
20. Romano, Y., Sesia, M., Candès, E.J.: Classification with valid and adaptive coverage (2020)
21. Rußwurm, M., Pelletier, C., Zollner, M., Lefèvre, S., Körner, M.: BreizhCrops: A time series dataset for crop type mapping. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **XLIII-B2-2020**, 1545–1551 (2020). <https://doi.org/10.5194/isprs-archives-XLIII-B2-2020-1545-2020>
22. Sainte Fare Garnot, V., Landrieu, L., Chehata, N.: Multi-modal temporal attention models for crop mapping from satellite time series. *ISPRS Journal of Photogrammetry and Remote Sensing* **187**, 294–305 (2022). <https://doi.org/10.1016/j.isprsjprs.2022.03.012>
23. Turkoglu, M.O., D’Aronco, S., Perich, G., Liebisch, F., Streit, C., Schindler, K., Wegner, J.D.: Crop mapping from image time series: Deep learning with multi-scale label hierarchies. *Remote Sensing of Environment* **264**, 112603 (2021). <https://doi.org/10.1016/j.rse.2021.112603>